QOSSAIN AWAIS

AI Engineer · Agentic Systems · Applied ML · Full-Stack Deployment

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AI engineer shipping production agentic and geospatial ML systems a self-built agent harness with cross-session memory and sandboxed execution, paired with deep-learning detection pipelines deployed on FastAPI/Docker/React.

Oxford NDORMS research collaborator and Springer Nature co-author, contributing to applied AI research within the first two years of a CS degree.

Built and deployed 6+ end-to-end systems spanning multi-agent orchestration, RAG, computer vision, and satellite data pipelines trained 40+ end users on production data workflows.

SKILLS

Agentic & AI Systems: Agentic Orchestration, Tool Calling, Context Window Management, Cross-Session Agent Memory, RAG, Vector Search (FAISS/Chroma), LLM APIs, Docker Sandboxing, Multi-Agent Systems, Eval Harnesses

Machine Learning & Vision: PyTorch, TensorFlow, Scikit-learn, Computer Vision, CNNs (ResNet, DenseNet, UNet, UNet++), Time-Series Forecasting, Transformer & Post-Transformer Architectures (JEPA), Hugging Face, Ollama

Backend & Cloud: Python, FastAPI, Flask, Django, React, TypeScript, Next.js, Docker, GCP, AWS, DigitalOcean, PostgreSQL, REST APIs, CI/CD, Full-Stack Deployment

Data & Geospatial: SQL, GIS & Remote Sensing, Google Earth Engine, Satellite Data (Sentinel-1/2, Landsat, EMIT, TROPOMI, ERA5, SMAP), Data Visualization, Power BI, Advanced Excel

EXPERIENCE

Research Collaborator
University of Oxford, NDORMS (Remote) *2025 – Present*

- Sustained Oxford-standard reproducibility on a medical-imaging AI research pipeline, as measured by supervisor sign-off under Dr. Usman Nazir, by building and maintaining applied data-science and model-development workflows in an international research setting.

Teaching Assistant Advanced Excel & Power BI
Knowledge Stream Bank of Punjab *Nov 2025 – Feb 2026*

- Upskilled 40+ senior banking professionals on production data workflows, as measured by improved assessment outcomes, by designing hands-on labs covering data cleaning, DAX, and Power BI dashboard reporting and providing targeted one-on-one support.

Research Assistant, Center for AI Research
Beaconhouse National University *Summer 2025*

- Delivered a 3-scope institutional carbon-emissions inventory (energy, transport, facilities) for BNU, as measured by inclusion in a peer-reviewed Springer Nature manuscript, by building automated data pipelines across energy and operational datasets.

SELECTED PROJECTS

Agentic Developer Harness *(Open Source — Upcoming Release)*

- Architecting a self-hosted agentic developer harness for autonomous multi-step coding/research task execution, targeting a public eval suite (task success rate, execution time) ahead of release, by implementing tool-calling orchestration, cross-session memory, and Docker-sandboxed execution environments with isolated resource limits.

Multi-Agent Autonomous Trading System

- Unified 3 independent signal sources (computer-vision, time-series, LLM-based macro agents) into a single adaptive trading signal, as measured by feedback-driven signal convergence across live data streams, by engineering a decision-fusion orchestrator over specialized Python/PyTorch agents.

Hybrid Methane Plume Detection System *(1st Place, Halshanas Hackathon (GDG BNU))*

- Achieved zero false positives across 10 negative-control sites and a year-long temporal scan (F1=0.766 classifier, IoU=0.543 segmenter), by building a hybrid deep-learning + physics-based verification pipeline (ResNet34, physics-informed UNet, TROPOMI/Sentinel-2/Landsat fusion) and deploying it via FastAPI/Docker on DigitalOcean with a React-Leaflet dashboard.

Pakistan Flood Early Warning System

- Reached AUC=0.93 flood-risk prediction validated against 2022 Pakistan flood data (33M affected), by fusing 4 satellite/climate sources (SMAP, ERA5, Sentinel-1, OpenWeather) across 17 districts into a real-time dashboard with pixel-level soil-moisture heatmaps via Google Earth Engine.

Virtual AI Doctor — MSK X-ray Co-pilot *(MVP, Oxford NDORMS)*

- Built an MSK X-ray triage MVP (DenseNet-style CNN, MURA dataset) pairing model inference with a RAG-based clinical knowledge layer, clinician-review safety gating, and a patient-facing 3D result interface (React/Three.js), under Oxford NDORMS supervision.

OpenAthena *(Founder)*

- Shipped an MVP autonomous intelligence platform for ambient fraud, risk, and AML detection, by designing and owning the end-to-end system architecture across internal/external threat-detection signals as founder.

Independent AI Architecture Research *(Self-Directed)*

- Investigating post-transformer backbone architectures (JEPA, state-space models) as an independent research track, targeting AGI applications in scientific discovery and grid/sustainability optimization.

EDUCATION

Beaconhouse National University — Bachelor of Computer Science, 2024 – Present

Superior College Pattoki — Intermediate in Computer Science, Completed

PUBLICATIONS & ACHIEVEMENTS

- "Plant-Based Indoor Air Purification in the Pakistani Context," Discover Atmosphere, Springer Nature — Co-Author (Published)
- "Multi-Sensor Methane Plume Detection from Space," Visualising Climate Conference — Author (Submitted)
- Selected as an Oxford NDORMS research collaborator in the first year of undergraduate study — an opportunity typically reserved for graduate researchers.